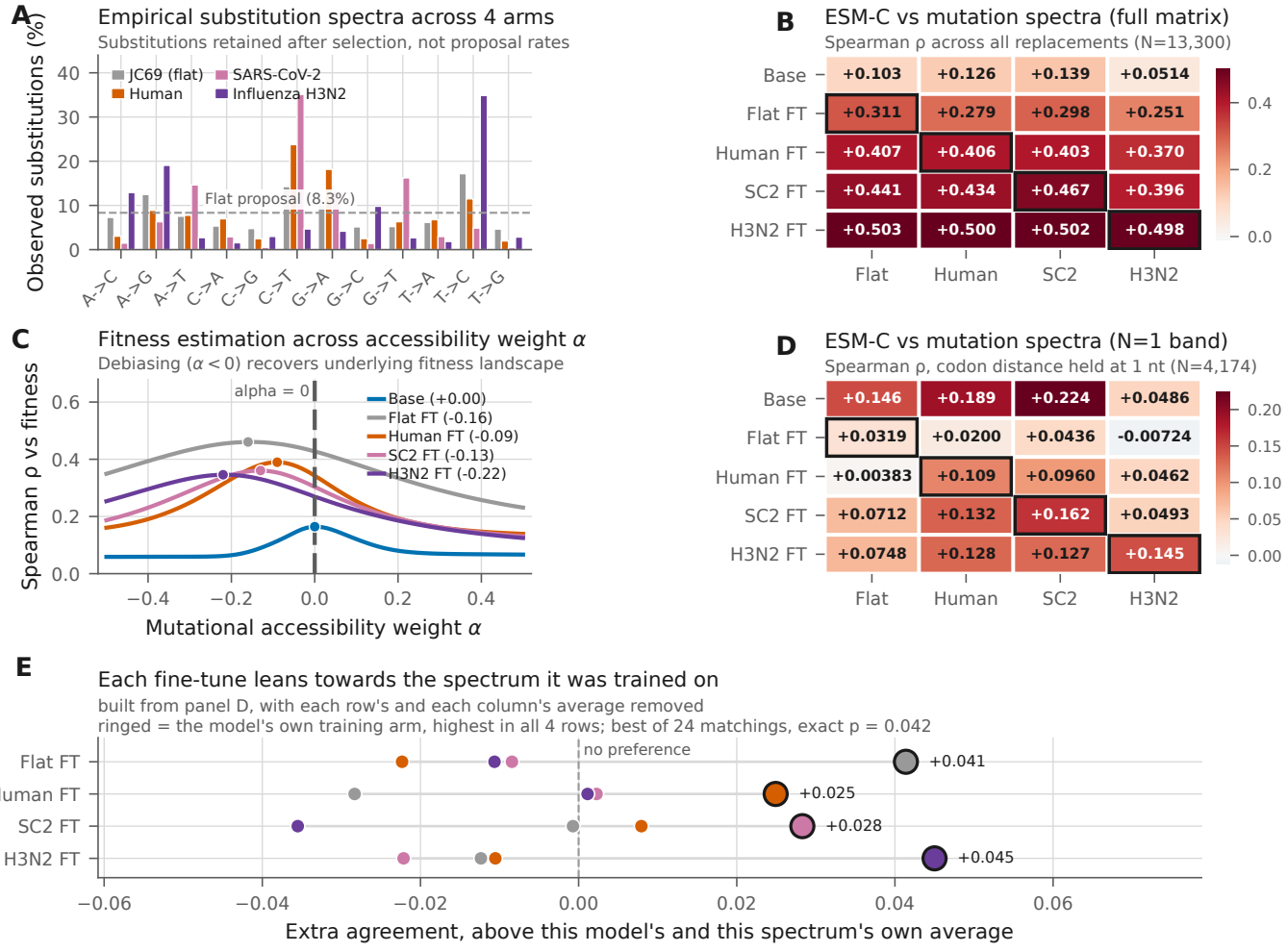
A
Empirical substitution spectra across 4 arms
Substitutions retained after selection, not proposal rates
Observed substitutions (%)
JC69 (flat)
SARS-CoV-2
Human
Influenza H3N2
Flat proposal (8.3%)
A->C
A->G
A->T
C->A
C->G
C->T
G->A
G->C
G->T
T->A
T->C
T->G
B
ESM-C vs mutation spectra (full matrix)
Spearman ρ across all replacements (N=13,300)
Base
Flat FT
Human FT
SC2 FT
H3N2 FT
+0.103
+0.126
+0.139
+0.0514
+0.311
+0.279
+0.298
+0.251
+0.407
+0.406
+0.403
+0.370
+0.441
+0.434
+0.467
+0.396
+0.503
+0.500
+0.502
+0.498
Flat
Human
SC2
H3N2
C
Fitness estimation across accessibility weight α
Debiasing (α < 0) recovers underlying fitness landscape
Spearman ρ vs fitness
alpha = 0
Base (+0.00)
Flat FT (-0.16)
Human FT (-0.09)
SC2 FT (-0.13)
H3N2 FT (-0.22)
Mutational accessibility weight α
D
ESM-C vs mutation spectra (N=1 band)
Spearman ρ, codon distance held at 1 nt (N=4,174)
Base
Flat FT
Human FT
SC2 FT
H3N2 FT
+0.146
+0.189
+0.224
+0.0486
+0.0319
+0.0200
+0.0436
-0.00724
+0.00383
+0.109
+0.0960
+0.0462
+0.0712
+0.132
+0.162
+0.0493
+0.0748
+0.128
+0.127
+0.145
Flat
Human
SC2
H3N2
E
Each fine-tune leans towards the spectrum it was trained on
built from panel D, with each row's and each column's average removed
ringed = the model's own training arm, highest in all 4 rows; best of 24 matchings, exact p = 0.042
no preference
Flat FT
Human FT
SC2 FT
H3N2 FT
+0.041
+0.025
+0.028
+0.045
Extra agreement, above this model's and this spectrum's own average